

Supplementary Material: Privacy Inference Analysis

This note provides an attack-surface summary and the mathematical derivations underlying the privacy discussion for the inner ADMM iterations of the proposed distributed framework. It characterizes what information can be inferred from the exchanged variables by the ISO, an individual wind farm, and an external observer, and clarifies which private quantities are not uniquely identifiable from the communication transcript.

1 Attack Surface Summary

For notational compactness in Table 1, boldface is omitted where the vector nature is clear, and we write

$$w_{s,m}^{\text{WF}} := \hat{\mathbf{w}}_{s,m}^{\text{WF}}, \quad w_m^{\text{ISO}} := \hat{\mathbf{w}}_m^{\text{ISO}}, \quad \bar{\tau}_m := \hat{\mathbf{w}}_m^{\text{glob}}, \quad \theta_s := \boldsymbol{\theta}_s, \quad x_m := \mathbf{x}_m, \quad y_m := \mathbf{y}_m, \quad z_m := \mathbf{z}_m.$$

2 Preliminaries

Fix one outer-loop master-problem iteration and consider the m -th ADMM iteration in the inner loop. Using the notation of the manuscript, define the ISO-side local variables

$$p := (x, y, \hat{w}^{\text{ISO}}), \quad p_{\text{co}} := \hat{w}^{\text{ISO}},$$

the WF-side local variables

$$q := (\hat{w}^{\text{WF}}, \theta), \quad q_{\text{co}} := \hat{w}^{\text{WF}},$$

and the global consensus variable

$$\bar{\tau} := \hat{w}^{\text{glob}}.$$

The ADMM subproblems and update equations are

$$p^{m+1} = \arg \min_{p \in P} \left(F(p) + (\eta_m^{\text{ISO}})^\top (p_{\text{co}} - \bar{\tau}_m) + \frac{\rho_m}{2} \|p_{\text{co}} - \bar{\tau}_m\|_2^2 \right), \quad (18)$$

$$q^{m+1} = \arg \min_{q \in Q} \left(G(q) + (\eta_m^{\text{WF}})^\top (q_{\text{co}} - \bar{\tau}_m) + \frac{\rho_m}{2} \|q_{\text{co}} - \bar{\tau}_m\|_2^2 \right), \quad (19)$$

$$\bar{\tau}_{m+1} = \frac{1}{2} \left(p_{\text{co}}^{m+1} + q_{\text{co}}^{m+1} + \frac{1}{\rho_m} (\eta_m^{\text{ISO}} + \eta_m^{\text{WF}}) \right), \quad (20)$$

$$\eta_{m+1}^{\text{ISO}} = \eta_m^{\text{ISO}} + \rho_m (p_{\text{co}}^{m+1} - \bar{\tau}_{m+1}), \quad (21)$$

$$\eta_{m+1}^{\text{WF}} = \eta_m^{\text{WF}} + \rho_m (q_{\text{co}}^{m+1} - \bar{\tau}_{m+1}). \quad (22)$$

Attacker	Directly known	Potentially inferable	Generally not uniquely inferable
ISO	x_m, y_m, z_m $w_{s,m}^{\text{WF}}, \bar{\tau}_{s,m}, \eta_{s,m}^{\text{WF}}, \rho_m$ $\lambda R_s(\theta_{s,m})$	$g_{s,m}^{\text{WF}} \in \partial \Psi_{s,m}(w_{s,m+1}^{\text{WF}})$	$H_s, \theta_{s,m}$ $\{p_s^{bl,j}(\cdot)\}_{j \in J_s}$ $\{(F_{i,s}, \tilde{w}_{i,s})\}_{i \in I_s}$
WF s	$H_s, \theta_{s,m}$ $\{p_s^{bl,j}(\cdot)\}_{j \in J_s}$ $\{(F_{i,s}, \tilde{w}_{i,s})\}_{i \in I_s}$ $w_{s,m}^{\text{WF}}, \eta_{s,m}^{\text{WF}}, \bar{\tau}_{s,m}, \rho_m$	$w_{s,m}^{\text{ISO}}, \eta_{s,m}^{\text{ISO}}$ $g_{s,m}^{\text{ISO}} \in \partial_s \Phi_m(w_{m+1}^{\text{ISO}})$	x_m, y_m, z_m, ξ_m^* $(A, B, c_1, c_2, E, G, K,$ $(\text{cuts}, \dots))$
External observer	$w_{s,m}^{\text{ISO}}, w_{s,m}^{\text{WF}}, \bar{\tau}_{s,m}$ $\eta_{s,m}^{\text{ISO}}, \eta_{s,m}^{\text{WF}}, \rho_m$ $\lambda R_s(\theta_{s,m})$	$g_{s,m}^{\text{WF}}, g_{s,m}^{\text{ISO}}$ g_m^{ISO}	$H_s, \theta_{s,m}$ $\{p_s^{bl,j}(\cdot)\}_{j \in J_s}$ $\{(F_{i,s}, \tilde{w}_{i,s})\}_{i \in I_s}$ x_m, y_m, z_m, ξ_m^* $(A, B, c_1, c_2, E, G, K,$ $(\text{cuts}, \dots))$

Table 1: Attack surface summary for the ADMM communication transcript.

For a single WF s , the second-layer meta-learner satisfies

$$\hat{w}_{i,s}^{\text{WF}} = \theta_{s,0} \mathbf{1} + \sum_{j \in J_s} \theta_{s,j} \hat{w}_{i,s}^{bl,j}, \quad \forall i \in I_s. \quad (14b)$$

Equivalently, in matrix form,

$$w_s = H_s \theta_s,$$

where $w_s := \hat{w}_s^{\text{WF}}$, the first column of H_s is the all-one vector, and the remaining columns are the precomputed base-learner outputs $\hat{w}_{i,s}^{bl,j}$. We assume H_s is elementwise positive.

3 Inference by the ISO

For WF s , the ISO directly knows

$$\{x_m, y_m, z_m, w_{s,m}^{\text{WF}}, \bar{\tau}_{s,m}, \eta_{s,m}^{\text{WF}}, \rho_m, \lambda R_s(\theta_{s,m})\}_m.$$

However, it does not directly know

$$F_{i,s}, \tilde{w}_{i,s}, p_s^{bl,j}(\cdot), \hat{w}_{i,s}^{bl,j}, H_s, \theta_{s,m}.$$

To characterize what the ISO can infer, define for each WF s the reduced value function

$$\Psi_{s,m}(w) := \min_{\theta_s: (w, \theta_s) \in Q_s} G_s(w, \theta_s), \quad (V1)$$

where Q_s is induced by $w = H_s \theta_s$, and G_s contains the local WF objective, including the regularization term $\lambda R_s(\theta_s)$. Then the WF subproblem is equivalent to

$$w_{s,m+1}^{\text{WF}} \in \arg \min_w \left\{ \Psi_{s,m}(w) + (\eta_{s,m}^{\text{WF}})^\top (w - \bar{\tau}_{s,m}) + \frac{\rho_m}{2} \|w - \bar{\tau}_{s,m}\|_2^2 \right\}. \quad (V2)$$

If $\Psi_{s,m}$ is a closed convex function, the first-order optimality condition gives

$$0 \in \partial\Psi_{s,m}(w_{s,m+1}^{\text{WF}}) + \eta_{s,m}^{\text{WF}} + \rho_m(w_{s,m+1}^{\text{WF}} - \bar{\tau}_{s,m}). \quad (\text{V3})$$

Therefore,

$$g_{s,m}^{\text{WF}} := -\eta_{s,m}^{\text{WF}} - \rho_m(w_{s,m+1}^{\text{WF}} - \bar{\tau}_{s,m}) \in \partial\Psi_{s,m}(w_{s,m+1}^{\text{WF}}). \quad (\text{ISO-1})$$

Since the ISO directly knows $w_{s,m+1}^{\text{WF}}$, $\eta_{s,m}^{\text{WF}}$, $\bar{\tau}_{s,m}$, and ρ_m , it can compute $g_{s,m}^{\text{WF}}$ exactly. Hence, the ISO can recover one exact subgradient of the WF reduced value function at the current forecast point.

Its key observations include

$$w_{s,m}^{\text{WF}} = H_s \theta_{s,m}, \quad \lambda R_s(\theta_{s,m}). \quad (\text{ISO-2})$$

Now consider any orthogonal transform O satisfying

$$O^\top O = I, \quad H'_s := H_s O \text{ remains elementwise positive.} \quad (\text{ISO-3})$$

Define

$$H'_s := H_s O, \quad \theta'_{s,m} := O^\top \theta_{s,m}. \quad (\text{ISO-4})$$

Then

$$H'_s \theta'_{s,m} = (H_s O)(O^\top \theta_{s,m}) = H_s \theta_{s,m} = w_{s,m}^{\text{WF}}. \quad (\text{ISO-5})$$

Moreover, for the Euclidean / Frobenius-type norm regularization used in this work,

$$R_s(\theta'_{s,m}) = R_s(O^\top \theta_{s,m}) = R_s(\theta_{s,m}). \quad (\text{ISO-6})$$

Hence, as long as a nontrivial transform O satisfying (ISO-3) exists, the same observable quantities can be generated by multiple different pairs $(H_s, \{\theta_{s,m}\}_m)$. Therefore, H_s and $\theta_{s,m}$ are not uniquely identifiable from the ISO's observations. Since the columns of H_s are precisely the base-learner outputs, the set

$$\{\hat{w}_{i,s}^{bl,j}\}$$

is not uniquely identifiable either. Because these outputs are generated through

$$\hat{w}_{i,s}^{bl,j} = p_s^{bl,j}(F_{i,s}),$$

the raw features $F_{i,s}$ and the private predictor internals $p_s^{bl,j}(\cdot)$ are even less likely to be uniquely reconstructed from the ADMM interaction alone.

4 Inference by a Wind Farm

WF s directly knows its own local quantities

$$H_s, \theta_{s,m}, w_{s,m}^{\text{WF}} = H_s \theta_{s,m}, \eta_{s,m}^{\text{WF}}, \bar{\tau}_{s,m}, \rho_m,$$

as well as the public algorithm structure. It does not know the ISO's complete UC / re-dispatch data, feasible region, or the private data and models of other WFs.

To determine what WF s can infer from the ADMM interaction, first derive the ISO-side copy and ISO-side dual. In the first OBCD run, the ADMM global variable and dual variables are initialized to zero, so

$$\eta_0^{\text{ISO}} + \eta_0^{\text{WF}} = 0. \quad (\text{I0})$$

Summing (21) and (22) yields

$$\eta_{m+1}^{\text{ISO}} + \eta_{m+1}^{\text{WF}} = \eta_m^{\text{ISO}} + \eta_m^{\text{WF}} + \rho_m(p_{\text{co}}^{m+1} + q_{\text{co}}^{m+1} - 2\bar{\tau}_{m+1}). \quad (\text{L1a})$$

Using (20),

$$2\bar{\tau}_{m+1} = p_{\text{co}}^{m+1} + q_{\text{co}}^{m+1} + \frac{1}{\rho_m}(\eta_m^{\text{ISO}} + \eta_m^{\text{WF}}). \quad (\text{L1b})$$

Substituting (L1b) into (L1a) gives

$$\eta_{m+1}^{\text{ISO}} + \eta_{m+1}^{\text{WF}} = 0, \quad \forall m \geq 0. \quad (\text{L1})$$

Therefore,

$$\eta_{s,m}^{\text{ISO}} = -\eta_{s,m}^{\text{WF}}, \quad \forall m \geq 0. \quad (\text{C1})$$

Substituting (L1) back into (20) further yields

$$\bar{\tau}_m = \frac{1}{2}(w_m^{\text{ISO}} + w_m^{\text{WF}}), \quad (\text{C2})$$

and hence, on the block corresponding to WF s ,

$$w_{s,m}^{\text{ISO}} = 2\bar{\tau}_{s,m} - w_{s,m}^{\text{WF}}. \quad (\text{C3})$$

Thus, WF s can algebraically reconstruct both the ISO-side forecast copy and the ISO-side dual on its own forecast block.

Define the ISO-side reduced value function

$$\Phi_m(w) := \min_{(x,y): (x,y,w) \in P_m} F_m(x, y, w), \quad (\text{V5})$$

where P_m is determined by the ISO operating constraints, the optimality cuts, and the fixed β -block in the OBCD procedure. Then the ISO subproblem is equivalent to

$$w_{m+1}^{\text{ISO}} \in \arg \min_w \left\{ \Phi_m(w) + (\eta_m^{\text{ISO}})^\top (w - \bar{\tau}_m) + \frac{\rho_m}{2} \|w - \bar{\tau}_m\|_2^2 \right\}. \quad (\text{V6})$$

If Φ_m is a closed convex function, then

$$0 \in \partial \Phi_m(w_{m+1}^{\text{ISO}}) + \eta_m^{\text{ISO}} + \rho_m(w_{m+1}^{\text{ISO}} - \bar{\tau}_m), \quad (\text{V7})$$

which implies

$$g_m^{\text{ISO}} := -\eta_m^{\text{ISO}} - \rho_m(w_{m+1}^{\text{ISO}} - \bar{\tau}_m) \in \partial \Phi_m(w_{m+1}^{\text{ISO}}). \quad (\text{V8})$$

Restricting to WF s 's forecast block gives

$$g_{s,m}^{\text{ISO}} = -\eta_{s,m}^{\text{ISO}} - \rho_m(w_{s,m+1}^{\text{ISO}} - \bar{\tau}_{s,m}). \quad (\text{WF-1})$$

Using (C1) and (C3), this can be rewritten as

$$g_{s,m}^{\text{ISO}} = \eta_{s,m}^{\text{WF}} - \rho_m(\bar{\tau}_{s,m} - w_{s,m+1}^{\text{WF}}) \in \partial_s \Phi_m(w_{m+1}^{\text{ISO}}). \quad (\text{WF-2})$$

Therefore, WF s can recover not only the ISO-side forecast copy and dual on its block, but also one exact marginal sensitivity of the ISO reduced value function on that block.

Under the simplified UC representation,

$$\min_{x,z} c_1^\top x + c_2^\top z \quad \text{s.t.} \quad Ax + Bz \leq d(\hat{w}), \quad (\text{WF-3})$$

where ξ denotes the dual variable associated with the constraints. If the binary structure is locally fixed and we consider the continuous relaxed value function

$$V_{\text{UC}}(w) := \min_x \{c^\top x : Ax \leq d(w)\}, \quad (\text{WF-4})$$

then, by the standard envelope / Danskin formula,

$$g_{\text{UC}}(w) \in -J_d(w)^\top \xi^*(w). \quad (\text{WF-5})$$

Hence, in this simplified local interpretation,

$$g_{s,m}^{\text{ISO}} \approx -J_{d,s}(w)^\top \xi_m^*, \quad (\text{WF-6})$$

where “ $\approx$ ” indicates the dominant part after ignoring the additional cut-penalty terms. Thus, WF s observes the projection of the ISO shadow-price information onto its own forecast block.

WF s still cannot uniquely recover the ISO’s full internal information. Even in the simplified setting above, WF s only observes

$$g_{s,m} = -J_{d,s}(w)^\top \xi_m^*. \quad (\text{WF-7})$$

Let $D_s := J_{d,s}(w)$. Then WF s only knows

$$D_s^\top \xi = -g_{s,m}. \quad (\text{WF-8})$$

If $\text{Null}(D_s^\top) \neq \{0\}$, then for any $v \in \text{Null}(D_s^\top)$,

$$D_s^\top (\xi + v) = D_s^\top \xi. \quad (\text{WF-9})$$

Therefore, ξ is generally not unique. In UC, the number of constraints is typically much larger than the forecast dimension, so this non-uniqueness is the norm rather than the exception. Moreover, the ADMM communication equations (20)–(22) involve only the forecast-side copies, consensus variable, dual variables, and penalty parameter; they do not explicitly contain

$$x_m, y_m, z_m, \xi_m^*.$$

Hence, WF s cannot algebraically recover the ISO’s internal primal variables from the communi-

cation equations alone. Because WF s only observes finitely many pairs

$$\{w_{s,m}^{\text{ISO}}, g_{s,m}^{\text{ISO}}\}_{m=1}^M,$$

there exist infinitely many different ISO-internal model data sets

$$(A, B, c_1, c_2, E, G, K, \text{cuts}, \dots)$$

that can induce the same value-function and subgradient information on those finitely many observed points. Therefore, the complete ISO UC / re-dispatch model data are not uniquely identifiable.

5 Inference by an External Observer

An external observer who can listen to all network messages but does not own either side's local data directly observes

$$\{w_{s,m}^{\text{ISO}}, w_{s,m}^{\text{WF}}, \bar{\tau}_{s,m}, \eta_{s,m}^{\text{ISO}}, \eta_{s,m}^{\text{WF}}, \rho_m, \lambda R_s(\theta_{s,m})\}_{s,m}.$$

Its inferential capability combines the previous two cases. In particular, it can recover the WF-side reduced value-function subgradients

$$g_{s,m}^{\text{WF}} := -\eta_{s,m}^{\text{WF}} - \rho_m(w_{s,m+1}^{\text{WF}} - \bar{\tau}_{s,m}) \in \partial \Psi_{s,m}(w_{s,m+1}^{\text{WF}}), \quad (\text{EXT-1})$$

the ISO-side block sensitivities

$$g_{s,m}^{\text{ISO}} := -\eta_{s,m}^{\text{ISO}} - \rho_m(w_{s,m+1}^{\text{ISO}} - \bar{\tau}_{s,m}) \in \partial_s \Phi_m(w_{m+1}^{\text{ISO}}), \quad (\text{EXT-2})$$

and, by aggregating all forecast blocks, the full-vector ISO subgradient

$$g_m^{\text{ISO}} := -\eta_m^{\text{ISO}} - \rho_m(w_{m+1}^{\text{ISO}} - \bar{\tau}_m) \in \partial \Phi_m(w_{m+1}^{\text{ISO}}). \quad (\text{EXT-3})$$

Thus, the core quantities recoverable by an external observer are still the forecast-side copies, the dual variables, and the first-order value-function information induced by them.

The same ambiguity argument used in the ISO case still applies: if there exists a nontrivial transform

$$O^\top O = I, \quad H'_s := H_s O \text{ remains elementwise positive}, \quad (\text{EXT-4})$$

and we define

$$H'_s := H_s O, \quad \theta'_{s,m} := O^\top \theta_{s,m}, \quad (\text{EXT-5})$$

then

$$H'_s \theta'_{s,m} = H_s \theta_{s,m} = w_{s,m}^{\text{WF}}, \quad (\text{EXT-6})$$

and, under the same norm-type regularization,

$$R_s(\theta'_{s,m}) = R_s(\theta_{s,m}). \quad (\text{EXT-7})$$

Therefore,

$$(H_s, \theta_{s,m}) \text{ is not uniquely identifiable.} \quad (\text{EXT-8})$$

Similarly, on the ISO side, the external observer generally recovers only projected dual information rather than the full dual vector. In the simplified local interpretation above,

$$g_m^{\text{ISO}} \approx -D^\top \xi_m^*, \quad (\text{EXT-9})$$

so the observer can at most identify

$$D^\top \xi_m^*, \quad (\text{EXT-10})$$

but not the full vector

$$\xi_m^*. \quad (\text{EXT-11})$$

Furthermore, the ADMM communication equations do not explicitly contain

$$x_m, \ y_m, \ z_m, \quad (\text{EXT-12})$$

so even an observer with access to all exchanged messages still cannot uniquely reconstruct the ISO's internal primal trajectory from the communication records alone.